What Should We Tell Our Students?

David J. Cox

So was the title of a recent blog written by Alvaro Lozano-Robledo on the site "What's New", edited, authored, and curated by Fields Medalist Terrence Tao. Lozano-Robledo raised one among an avalanche of questions, reactions, and emotional responses to recent work published by OpenAI. The frontier lab had used their latest artificial intelligence (AI) system to provide proofs solving a mathematical problem previously considered career-level difficult, if not utterly intractable. The time seemed to come for AI systems to forever surpass humans in yet another intellectual realm previously considered uniquely human. Mathematics had fallen.

Those learned about this space will recognize the story. Chess. Go. Basic writing. Visual imagery. Graphic design. Music. Software development. Now mathematics. In their sights are physics, engineering, wet labs, education. In each case, AI systems were developed, their outputs initially met with humor at the simplicity and error rate, and with claims that the AI system is unlikely to ever be able to do what a human could do. Then, one day, those considered pinnacle performers in their intellectual domain were challenged to a duel. Sometimes directly, sometimes through less capable colleagues. In each case the right to claim top performer was taken from humans and awarded to AI.

Mathematics had fallen.

One among an avalanche

Different people in the different industries responded differently to the death of human dominance. Some with enthusiasm at the skills they now wield in the palm of their hand. Others with anxiety at an uncertain imagined future with robot overlords. Others absorbed the technology into their day-to-day work and accept the transformation driven by survival contingencies. Still others react with anger and fear at the frontier labs' open disregard for the fields, no professions, *no livelihoods!* that were being upended. *People's* lives upended. Overnight. Each no longer the best at that thing they worked years to develop expertise and fluency to perform. Now, for cents on the dollar, anyone can access and bear fruit using the top 1% of skills in that profession.

Exhilarating for many. Existential for a few.

One among an avalanche

That people responded differently in similar situations is unsurprising. So it goes with the infinite combinations of cause-effect behavior-environment relations that accumulate throughout a lifetime.

But, though the specific responses may differ, the stream of questions seem similar. What do I do next? What new skills do I need? What if I do nothing? What can I do now that I couldn't before? What do I tell my colleagues? How are they using it? How can I maintain my hard-earned skills? What if those skills become obsolete? What can I learn now that I couldn't before? How creative am I? Can I adapt? Can everything I do be reduced to what an AI system can do? What are uniquely human skills? What jobs require those? Can I pivot? How well do those pay? Will I be able to continue to make a living? Will I be able to support my family? How will I sustain the energy to continuously adapt and change for the remaining 30+ years of my career if AI advances continue at their current pace?

One among an avalanche

The Gambler said you can know someone's cards by the way they hold their eyes. Some mathematicians sought to hold the gaze of frontier labs. "Recognize us as humans", they requested. "It's not that the intractable problem had been solved, but the way you went about it!" It was a brute force solution. It used ideas humans had earlier input into the system. This was theft, not orig-

inality, not something to be celebrated. Plus, mathematics is not just proof building. Mathematics involves bringing the community of *people* along. Help the rest understand how it works. Help the rest understand how to use it. Help the rest of us get to where you are. Help "*people* to understand and think more clearly and effectively about mathematics", says Thurston. For mathematics—all sciences—are community affairs. Human affairs. Be a good citizen of the community if you wish to join. And stop what you're doing if you can't participate properly in our community according to our values!

The calls grew near legion and ended in petition. An open letter from mathematicians against AI use for mathematics. It was a "severe misalignment".

What was OpenAI's response to the pleas to participate cordially? Publication of 700+ more proofs and papers. Overnight. In an obscure location called a GitHub repo and titled simply, "math". No fanfare. A step on the neck.

Exhilarating for some. Existential for a few.

What should we tell our students?

Mathematics had fallen.

Though the specifics may differ, the behavior change driven by the contingencies for all in the path sing familiar to behavior analysts. Adaptation, selection, behavior change. We ought to be model citizens here, no? We should know best how to navigate these waters, no? We should know best how to help others. We should know best how to help ourselves. The control of operant variability is in our blood. The acquisition of new behavior is the jam we have with our breakfast toast, two eggs over-easy, and black coffee. Time to generalize our skills and to help others do the same. We got this, right?

The dominoes are falling. In their sights are physics, engineering, wet labs, education.

We're somewhere down that line.

One among an avalanche

So, what should we tell others, ourselves, our students? What sage advice might the science of behavior have to offer? The same advice offered since an-

cient times and refined over the past century through experimentation: To live. Behave. Contact the contingencies. Be brave enough to suck at something new. Adaptation occurs via selection acting on variation. “Regard no practice as immutable. Change and be ready to change again. Accept no eternal verity. Experiment.”, says Skinner.

Chess. Go. Writing. Visual imagery. Graphic design. Music. Software development. All have walked the path before us. “Read, learn, work it up, go to the literature”, says Didion. Each show the change forthcoming is transformation, not death. And each person will learn what that transformed repertoire must look like for them. Adaptation. Survival.

Now mathematics.

That transformation is underway. Watch. Learn.

In their sights are physics, engineering, wet labs, education. The dominoes are falling. We’re somewhere down that line.

The lesson of the snowflake in the avalanche is simple. All was calm before. All is calm after. And, though the violent change comes suddenly, we’re all in this together.

One among an avalanche

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Notes and Sources

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